

Konstantinos Christopher Tsiolis

kc.tsiolis@mail.utoronto.ca | [Website](#) | [LinkedIn](#) | [GitHub](#)

PROFILE

I am a PhD candidate in the Department of Statistical Sciences at the University of Toronto and the Vector Institute, supervised by Murat Erdogdu. My research is on theoretical aspects of machine learning, with a focus on feature learning and scaling laws.

EDUCATION

Doctor of Philosophy (PhD), Statistics <i>University of Toronto</i> - Supervisor: Murat Erdogdu - Cumulative GPA: 4.0/4.0	2023 – Present Toronto, ON
Master of Science (MSc), Mathematics and Statistics <i>McGill University</i> - Supervisor: Adam Oberman - Cumulative GPA: 4.0/4.0 - Thesis: Contrastive Learning as Kernel Approximation [Link]	2022 – 2023 Montreal, QC
Bachelor of Science (BSc), Mathematics and Computer Science (Honours) <i>McGill University</i> - Cumulative GPA: 3.99/4.0 - Research supervisors: Jackie Cheung (2019-2020), Adam Oberman (2021) - Implemented and documented the training word embeddings with various context window sizes [Summary] - Wrote a paper on ensembling pre-trained embeddings from various datasets and algorithms [Paper] - Developed, implemented, and tested an approach for knowledge distillation of self-supervised models [Code]	2018 – 2021 Montreal, QC

PAPERS

- **Konstantinos Christopher Tsiolis**, Denny Wu, Christos Trampoulidis, and Murat A. Erdogdu. SGD in Multiclass Logistic Regression: Sequential Learning and Scaling Laws. arXiv preprint arXiv:2609.07868, 2026. [\[Paper\]](#)
- **Konstantinos Christopher Tsiolis**, Alireza Mousavi-Hosseini, and Murat A. Erdogdu. From Information to Generative Exponent: Learning Rate Induces Phase Transitions in SGD. In *Proceedings of the 39th International Conference on Neural Information Processing Systems* (NeurIPS). 2025. [\[Paper\]](#) [\[Code\]](#)
- Jingyi He, **KC Tsiolis**, Kian Kenyon-Dean, and Jackie Chi Kit Cheung. Learning efficient task-specific meta-embeddings with word prisms. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 1229–1241. International Committee on Computational Linguistics, 2020. [\[Paper\]](#) [\[Code\]](#)

AWARDS

NSERC Postgraduate Scholarship – Doctoral (PGS D) (120,000 CAD)	2024 – 2027
Faculty of Arts and Science Top (FAST) Doctoral Fellowship (20,000 CAD)	2023 – 2027
Department of Statistical Sciences Student Leadership Award (1,000 CAD)	2025
Queen Elizabeth II Graduate Scholarship in Science and Technology (QEII-GSST) (15,000 CAD)	2023
FRQNT Master’s (B1X) Research Scholarship (17,500 CAD) Application ranked 1st out of 17 in Mathematics category	2022
NSERC Canada Graduate Scholarships - Master’s Program (CGS M) (17,500 CAD)	2022 – 2023
IVADO Undergraduate Introduction to Research Scholarship (6,500 CAD)	2021
NSERC Undergraduate Student Research Award (4,500 CAD)	2020
McGill University Undergraduate Awards	2018 – 2021

- John Mahon Scholarship (12,000 CAD)
- Alan Shaver Scholarship (2,000 CAD)
- Anne Molson Scholarship (1,200 CAD)
- Emily Ross Crawford Scholarship (1,000 CAD)
- Wing Hing Chan Scholarship in Science (500 CAD)

LEADERSHIP AND SERVICE

Mentorship

- **STA492, University of Toronto (Fall 2026):** Mentoring undergraduate students in AI-assisted research development. Funded by a Pedagogical Innovation and Experimentation grant, Department of Statistical Sciences.
- **University of Toronto Statistical Sciences Directed Reading Program (Winter 2026):** Weekly meetings with 4 undergraduate students covering graduate-level readings on optimal transport, mean-field neural networks, and lazy vs. rich learning regimes
- **Academic Lecturer at the McGill Artificial Intelligence Society (2020 – 2021):** Giving 4 two-hour lectures for a student-run machine learning course

Conference Organizer and Reviewer

- Reviewer for *NeurIPS* (2026), *COLT* (2026), *Workshop on High Dimensional Learning Dynamics* (2023 – 2026)
- Member of the Local Organizing Committee for *The Fast and Curious 2: MCMC in Action* (2025)

President of the University of Toronto Statistics Graduate Student Union (SGSU)

- Elected representative of the PhD and MSc students in the department for the academic year 2024 – 2025
- Lead organizer of Graduate Student Research Day

TEACHING EXPERIENCE

Teaching Assistant

- **At the University of Toronto (2023 – Present):** STA2311 and STA2312 (Advanced Computational Methods for Statistics I and II, *PhD Course*), STA347 (Probability), STA314 (Statistical Methods for Machine Learning I), STA261 (Probability and Statistics II), STA221 (The Practice of Statistics II), STA130 (An Introduction to Statistical Reasoning and Data Science)
- **At McGill University (2021 – 2022):** MATH 222 (Calculus 3), COMP 251 (Algorithms and Data Structures)

SKILLS

Programming Languages: Python, R, Java, Bash, C, MATLAB, JavaScript, OCaml

Libraries: PyTorch, NumPy, JAX, scikit-learn, pandas, Matplotlib, tidyverse

Tools: Git, LaTeX, Jupyter, VS Code